

Dynamic comparison of Monod and substrate inhibition models for recombinant lipase production in a methanol fed batch bioreactor

CL 420, Introduction to Biochemical Engineering

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1. Motivation and objective

Upstream bioprocess design requires a connection between microbial kinetics and reactor operation. The CL 420 project instructions identify growth and product kinetics, material balances, oxygen demand, and bioreactor analysis as core engineering requirements [3, 4]. These requirements are addressed through a dynamic study of recombinant *Rhizopus oryzae* lipase production by *Pichia pastoris* during methanol induction.

The objective is to determine how the choice of substrate kinetics affects predicted biomass, residual methanol, product activity, productivity, and oxygen demand under identical fed batch operating conditions. The study evaluates a Monod formulation against a substrate inhibition formulation.

The kinetic parameters, initial conditions, and operating strategies were obtained from the published work of Ponte et al. [1, 2]. The Monod formulation uses the same maximum rates and saturation constants as the reported model, while omitting the inhibition term. This provides a direct comparison between the two kinetic assumptions.

2. Reference process and kinetic model

The modelled process uses a methanol-utilizing *Pichia pastoris* strain expressing *Rhizopus oryzae* lipase under the methanol-induced P_{AOX1} promoter. The initial conditions are $X_0 = 27 \text{ g L}^{-1}$, $P_0 = 50 \text{ U mL}^{-1}$, $V_0 = 2 \text{ L}$, and $S_0 = 1 \text{ g L}^{-1}$ for predefined feed profiles. The operating strategies are constant feed, constant growth, linearly increasing growth, and constant substrate concentration [2].

The reported non-monotonic substrate relationship for growth and product formation is

$$q_i(S) = q_{\max,i} \frac{S}{K_{S,i} + S + S^2/K_{I,i}}, \quad (1)$$

where i represents either biomass growth or product formation. The corresponding Monod relationship used for comparison is

$$q_i(S) = q_{\max,i} \frac{S}{K_{S,i} + S}. \quad (2)$$

The substrate inhibition relationship gives a maximum growth rate near 1.9 g L^{-1} and a maximum product formation rate near 3.5 g L^{-1} . The Monod relationship increases towards saturation and does not represent this high-substrate decline.

Table 1: Parameters used in both model formulations. Values were adopted from Ponte et al. [1, 2].

Parameter	Value	Unit	Parameter	Value	Unit
$\mu_{\max}$	0.069	h^{-1}	$q_{\max,P}$	2200	$\text{U g}^{-1} \text{ h}^{-1}$
$K_{S,X}$	0.50	g L^{-1}	$K_{S,P}$	10.0	g L^{-1}
$K_{I,X}$	7.5	g L^{-1}	$K_{I,P}$	1.2	g L^{-1}
$Y_{S/X}$	5.14	g g^{-1}	$m_{S/X}$	0.013	$\text{g g}^{-1} \text{ h}^{-1}$
$Y_{O_2/X}$	0.206	mol g^{-1}	$m_{O_2/X}$	6.1×10^{-4}	$\text{mol g}^{-1} \text{ h}^{-1}$

The uptake relations are

$$q_S = Y_{S/X} \mu + m_{S/X}, \quad q_{O_2} = Y_{O_2/X} \mu + m_{O_2/X}. \quad (3)$$

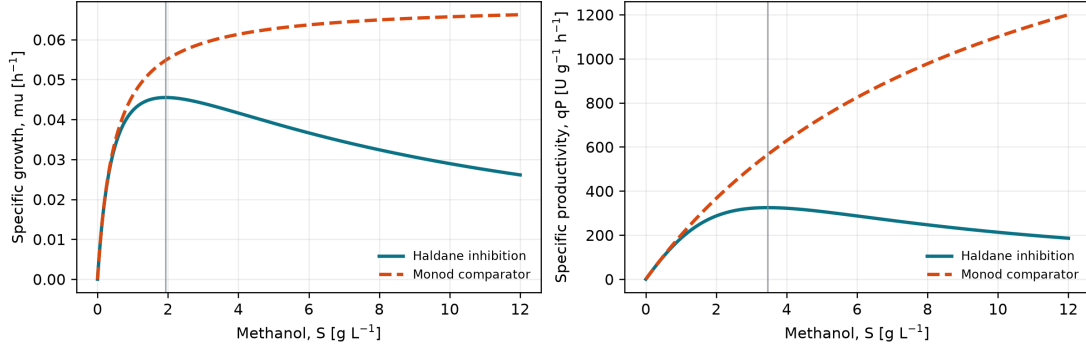


Figure 1: Specific growth and product formation laws used in the simulation. The vertical markers show the reported critical substrate concentrations for the inhibition branch.

3. Four state simulation method

The dynamic state vector is $[V, XV, SV, PV]$, where XV is biomass amount, SV is methanol amount, and PV is total product activity. With a pure methanol feed approximation, the balances are

$$\begin{aligned}\frac{dV}{dt} &= \frac{F}{\rho_{\text{methanol}}}, \\ \frac{d(XV)}{dt} &= \mu XV, \\ \frac{d(SV)}{dt} &= F - q_S XV, \\ \frac{d(PV)}{dt} &= q_P XV.\end{aligned}\tag{4}$$

The simulation uses $\rho_{\text{methanol}} = 792 \text{ g L}^{-1}$. Concentrations are recovered by dividing each amount by V . This amount formulation makes the feed and consumption terms explicit and avoids numerical ambiguity when the working volume changes.

Four methanol feeding strategies reported by Ponte et al. [1, 2] are considered. CF uses $F = 14 \text{ g h}^{-1}$, CM targets $\mu = 0.03 \text{ h}^{-1}$, LM increases the target growth rate from 0.01 to 0.04 h^{-1} , and CS holds $S = 2 \text{ g L}^{-1}$. The CS case is implemented using the algebraic feed balance. The published sources describe an adaptive predictive PI controller but do not specify all tuned controller gains, so the ideal setpoint balance is used for this four-state analysis.

The ODEs are integrated with classical fourth order Runge Kutta at $\Delta t = 0.01 \text{ h}$. The process stops at the first of $X = 55 \text{ g L}^{-1}$, $V = 5 \text{ L}$, $S = 10 \text{ g L}^{-1}$, or $t = 120 \text{ h}$, matching the operating limits used in the source study. The CS result was repeated with $\Delta t = 0.005 \text{ h}$ as a numerical convergence check.

Oxygen demand is derived after solving the four states. The total oxygen uptake rate is

$$\text{OUR} = q_{O_2} XV, \quad \text{OUR}_V = q_{O_2} X,\tag{5}$$

where OUR is in mol h^{-1} and OUR_V is in $\text{mol L}^{-1} \text{ h}^{-1}$. These quantities represent cellular oxygen demand. Oxygen transfer capacity is not evaluated in the present analysis.

4. Simulation results

Table 2 reports the simulated values at the first process limit. The substrate inhibition and Monod formulations use the same initial conditions, feed definitions, and process limits. The results are numerical predictions generated by the present implementation.

Figure 1 shows the difference between the two kinetic formulations. Monod kinetics increases with methanol and approaches saturation, whereas the substrate inhibition formulation reaches a maximum and then declines. The growth and product peaks occur at different substrate concentrations, so a feed that supports growth is not automatically optimal for product formation.

The CS strategy gives the highest performance in both formulations. At $S = 2 \text{ g L}^{-1}$, the substrate inhibition model predicts 255.01 U mL^{-1} , while the Monod formulation predicts 266.18 U mL^{-1} because it excludes substrate inhibition.

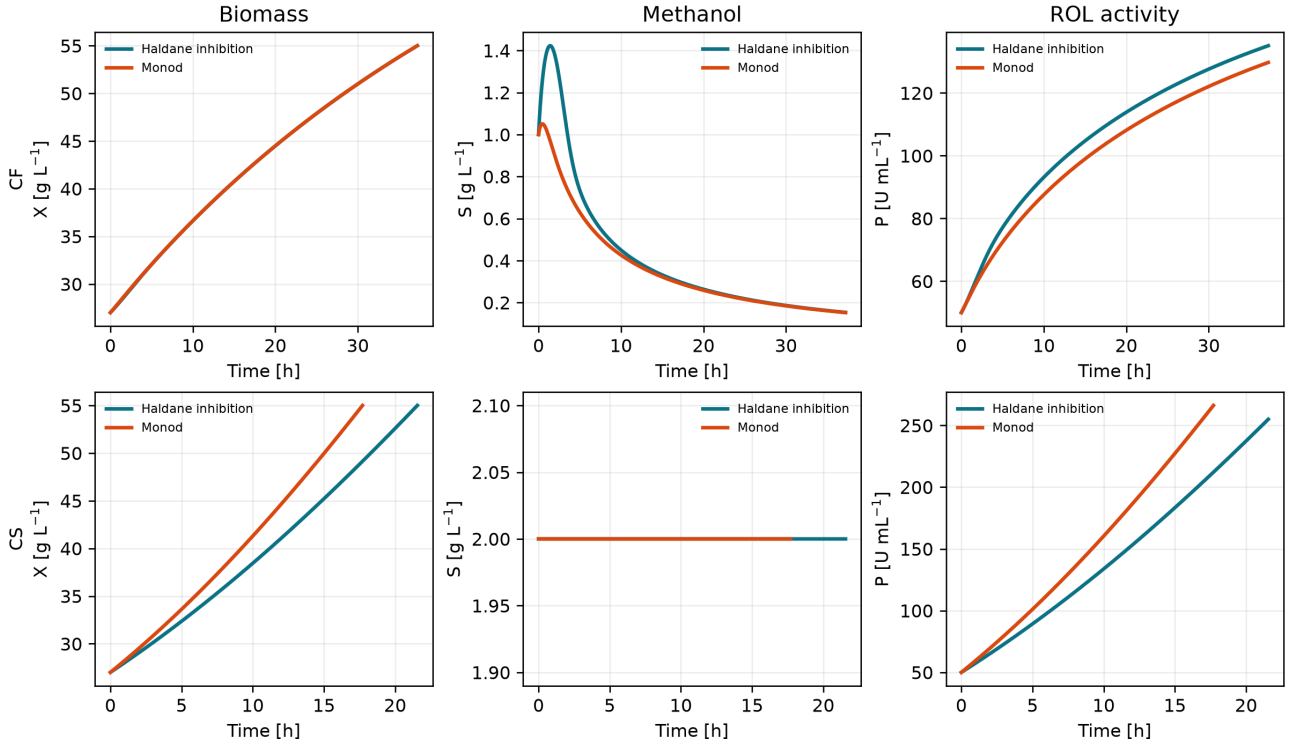


Figure 2: Four state comparison for constant feed CF and constant substrate CS. The same initial conditions and feed definitions are used in both branches.

Table 2: Computed performance for the four standard strategies. Q_P is the increase in total activity divided by final volume and process time. OUR_{max} is total oxygen demand.

Model	Strategy	t_f h	P_f $U \text{ mL}^{-1}$	Q_P $U \text{ L}^{-1} \text{ h}^{-1}$	OUR_{max} mol h^{-1}	Stop condition
Inhibition	CF	37.22	135.08	2619	0.589	X_{max}
Inhibition	CM	32.86	134.64	2945	0.986	X_{max}
Inhibition	LM	58.12	114.35	1328	0.838	X_{max}
Inhibition	CS	21.56	255.01	10057	1.437	X_{max}
Monod	CF	37.22	129.75	2475	0.576	X_{max}
Monod	CM	32.85	131.98	2865	0.986	X_{max}
Monod	LM	58.12	113.73	1318	0.838	X_{max}
Monod	CS	17.70	266.18	12870	1.719	X_{max}

5. Conclusion and limitations

This study presents a reproducible comparison of two substrate kinetic formulations in a four-state fed batch model. The parameter values and operating conditions were obtained from published work, while the dynamic simulations and performance calculations were carried out in the present implementation. The Monod formulation changes one modelling assumption while keeping the feed strategy, initial state, numerical solver, and process limits unchanged. The results show why substrate inhibition matters in bioreactor operation. It changes the predicted productivity and prevents a high residual methanol concentration from being treated as universally beneficial. The derived oxygen calculation adds a practical reactor design consequence: the best product strategy also creates the largest oxygen demand and therefore needs an oxygen transfer check in the final project.

The main limitation is scope. The source model also discusses oxygen and carbon dioxide balances, gas transfer, and an adaptive predictive PI controller. The present analysis keeps oxygen as a derived demand and does not include a full OTR or $k_L a$ design. It also does not reproduce the unpublished controller tuning parameters. These extensions can be added in a subsequent design stage.

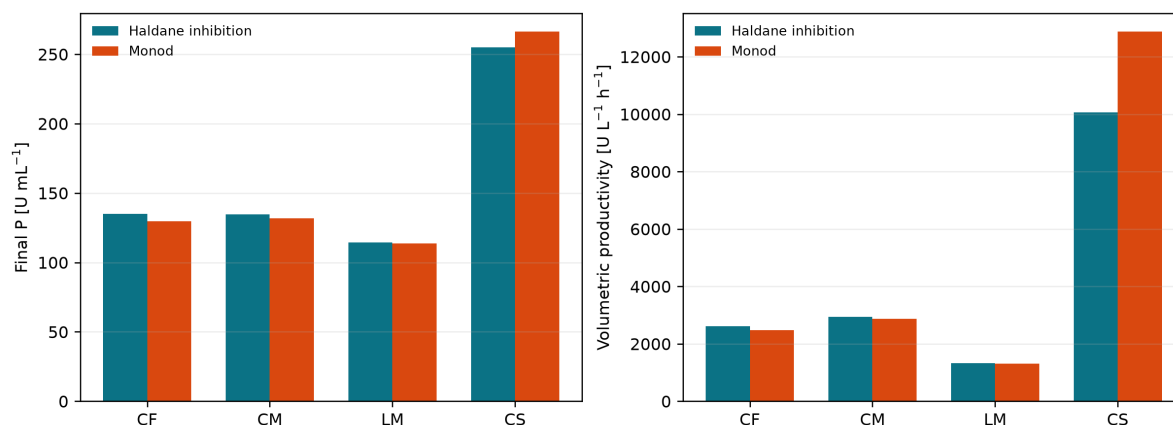


Figure 3: Final product activity and volumetric productivity for all four strategies. The two model branches use identical process limits.

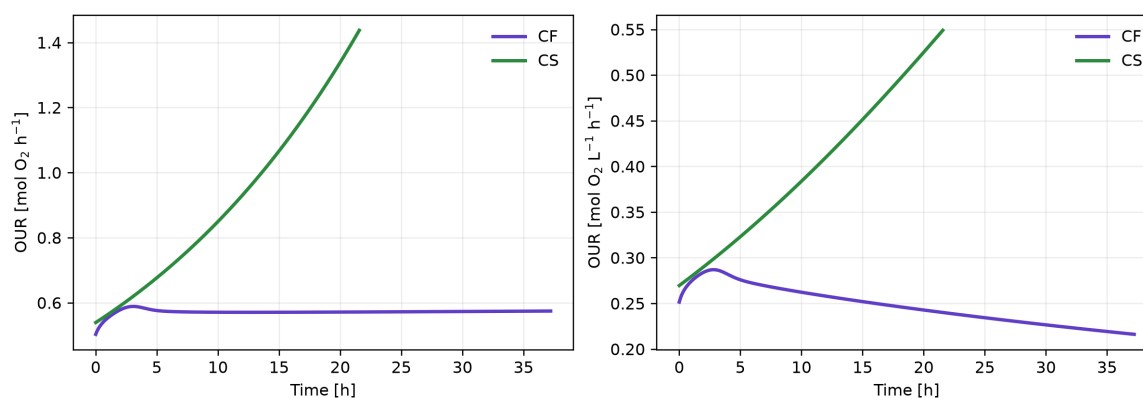


Figure 4: Derived oxygen demand for the inhibition branch. CS gives faster biomass and product formation, but it also creates the higher instantaneous oxygen requirement.

References

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